FENNO-AURUM
Nordic XRF Innovations



INDUSTRIAL PROCESS ANALYZER

ON-LINE CONVEYOR XRF ANALYZER

Real-time, non-contact and non-nuclear elemental grade control on the belt

BALTIC SCIENTIFIC INSTRUMENTS

DESIGNED & MANUFACTURED
Baltic Scientific Instruments
Riga, Latvia

FENNO-AURUM
Nordic XRF Innovations

GLOBALLY DISTRIBUTED BY
Vocto Technologies Pvt Ltd . India
Fenno-Aurum Oy . Finland

On-line Conveyor XRF Analyzer

Real-time, non-contact and non-nuclear elemental grade control directly on the belt

KEY BENEFITS

- Continuous, real-time grade measurement enables immediate blending and beneficiation decisions
- Non-contact, on-belt sensing eliminates manual sampling delays and laboratory turnaround time
- Ruggedized for harsh plant environments — dust, vibration, and continuous duty cycles
- Reduces reliance on off-line laboratory analysis, cutting grade-control costs
- Direct integration with plant SCADA/DCS for closed-loop process control
- It is a non nuclear device without the burden of nuclear licensing, safety and high costs.

HOW IT WORKS

The On-line conveyor XRF analyzer identifies and measures elemental and mineral concentrations directly on the conveyor belt, detecting elements from sulphur (Z=16) through uranium (Z=92). The analyzer head is mounted over or across the belt, irradiating the material stream and detecting characteristic X-ray fluorescence using a Silicon Drift Detector (SDD).

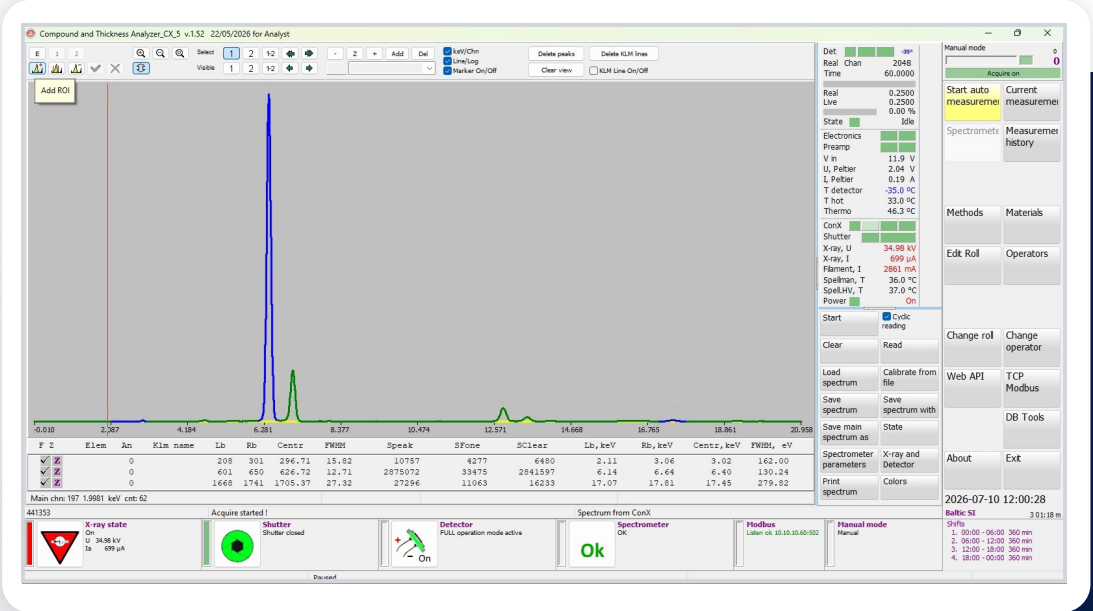


Fig. 1. Con-X analyzer software — live spectrum acquisition and elemental readout (Fe Kα / Kβ shown)

TECHNICAL SPECIFICATIONS

Detector type	Silicon Drift Detector (SDD)
Excitation source	Sealed X-ray tube, application-dependent
Elements measured	Sulphur (Z=16) through Uranium (Z=92)*
Measurement range	Configurable per ore matrix
Accuracy	Comparable to laboratory XRF after calibration against CRMs
Response / update time	Continuous, with rolling averages
Belt width compatibility	Configurable scanner geometry; cross-belt or over-belt
Ingress protection	IP54 (analyzer head and control/communication cabinet)
Operating temperature	-20°C to +45°C
Power supply	24V DC / 230V AC, plant-supplied
Power consumption	130 W (average power consumption is lower)
Power backup	UPS strongly recommended — not part of the Con-X system; must be provided by the customer at site
Output interfaces	Modbus TCP/RTU, OPC-UA
Mounting	Over-belt scanner or cross-belt frame; minimal civil works
Dimensions (L × B × H)	890 × 275 × 240 mm

*** Elements up to Arsenic (Z=33) are detected by their K-series lines; from Selenium (Z=34) onward, detection is by K- or L-series lines.**

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Measurable Element Range

Sulphur (Z=16) through Uranium (Z=92)

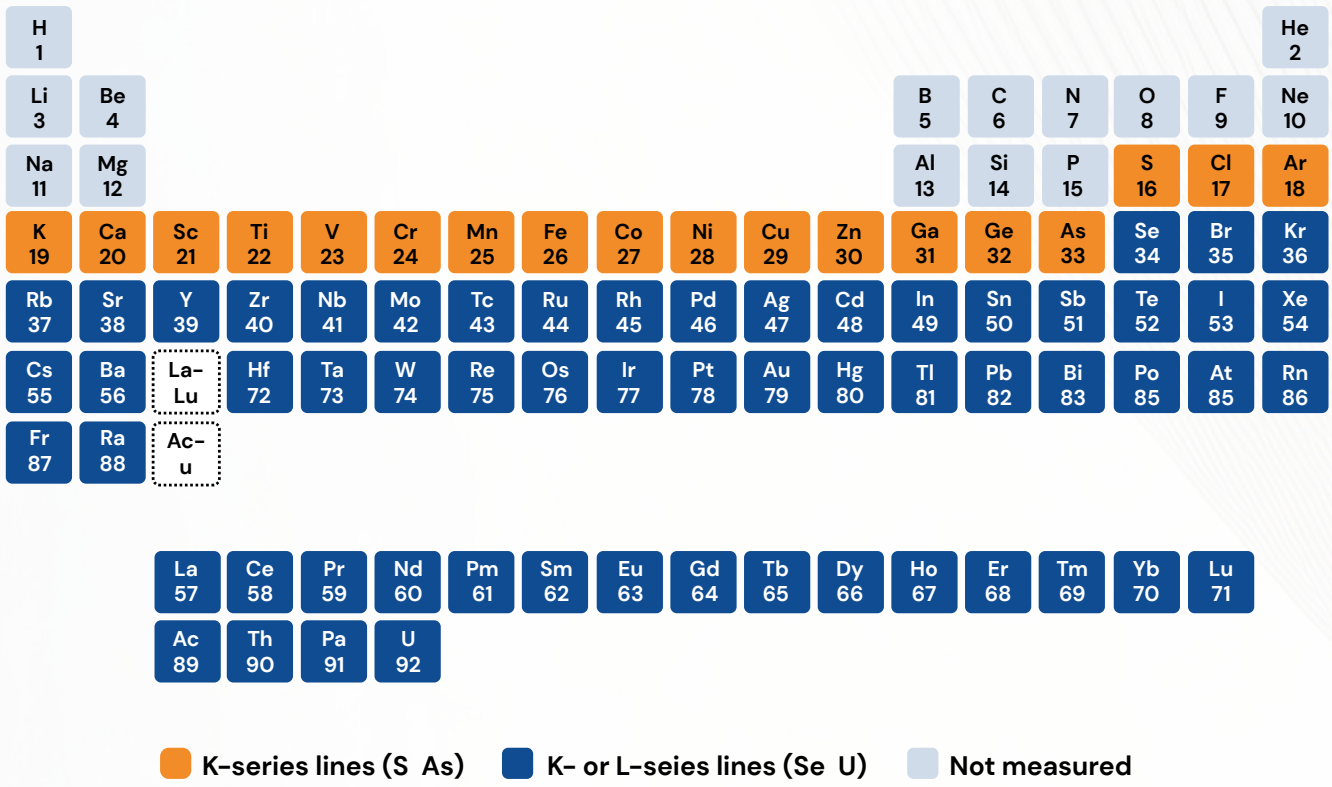


Fig. 2. Elemental detection range of the On-line conveyor XRF analyzer

PLANT INTEGRATION

- Direct feed into SCADA, DCS, or blending optimization software via Modbus / OPC-UA
- Configurable alarms and set-points for off-grade material diversion
- Historical trend logging for quality assurance and reporting

APPLICATIONS

- Iron ore
- Ferrochrome
- Manganese ore
- Other base metal ores
- Quartz sand (Fe content)
- Slag of various types
- Coatings on metal bulk material

FASTER ROI AND SAVINGS

- Savings in raw materials used
- Faster analysis, saving valuable plant time
- Increased plant productivity

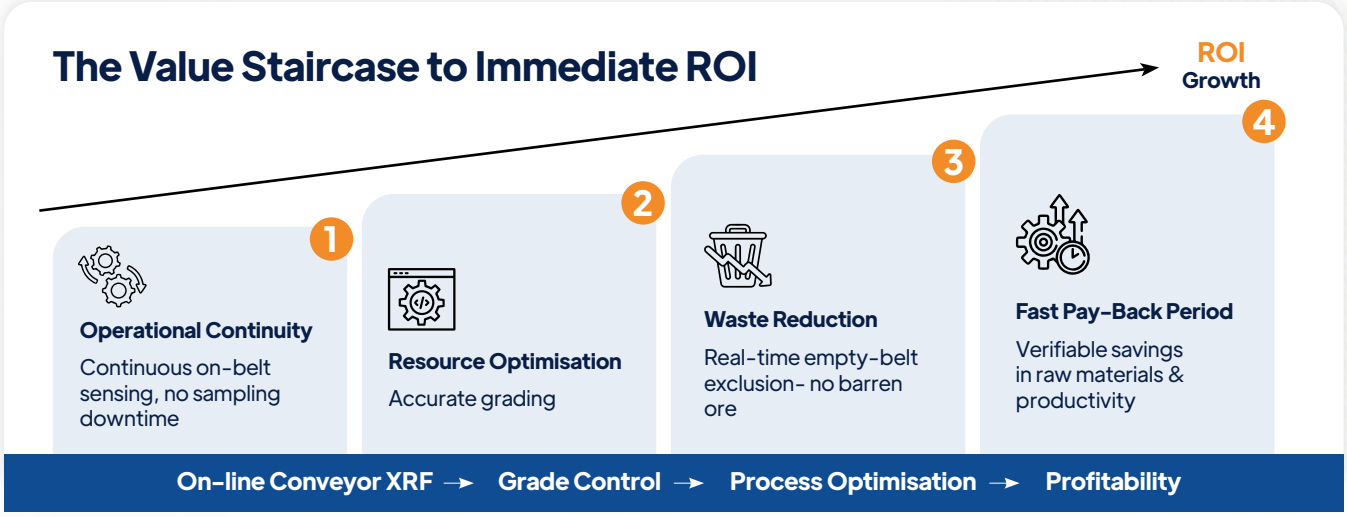


Fig. 3. The value staircase to immediate ROI

INSTALLATION CONSIDERATIONS

- Belt material level distance from the analyzer window should fall within the predetermined range
- Empty-belt readings are automatically excluded by the Con-X empty-belt algorithm; even so, installation should minimize belt vibration (unit weight approx. 45 kg)

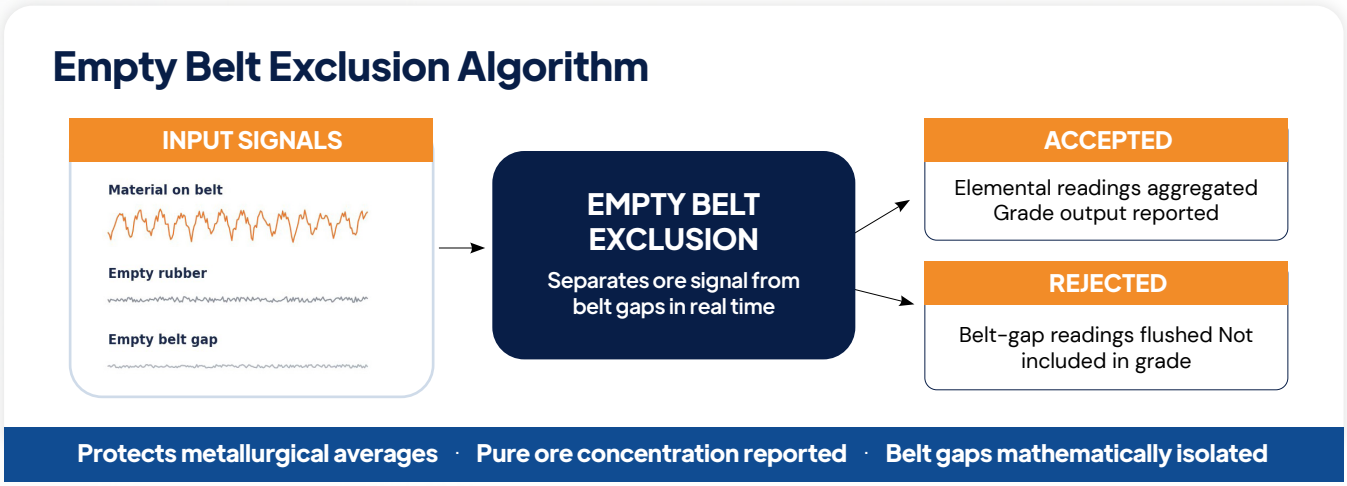


Fig. 4. Empty-belt exclusion algorithm: belt-gap readings are flushed so the reported grade reflects pure ore concentration

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- Multiple suspension options are available: frame, crane, cart, and ski mount
- Installation should avoid close proximity to EMI sources such as motors
- The analyzer reports an average result over the predetermined measurement time for material moving on the belt
- Statistical repeatability depends on standard XRF factors — grain size, distance to material surface, measurement time, and signal attenuation through material layers; periodic cleaning of the analyzer is recommended
- Calibration is performed using the customer's known reference samples

BUILT FOR HARSH ENVIRONMENTS

Uncompromising Stability in Extreme Environments

Three countermeasures protecting your measurement in the harshest plant conditions

Complete Environmental Sealing

Maintains precision even in extreme dust conditions — standard to raw-material processing

Thermal Regulation Core

Sustains accuracy across drastic temperature swings, both low and high

Humidity Isolation

Prevents condensation and sensor degradation for continuous 24/7 operation

Precision Maintained · 24/7 Availability · No Recalibration Downtime

Fig. 5. Environmental sealing, thermal regulation, and humidity isolation sustain accuracy in continuous industrial duty



Fig. 6. Con-X control and communication cabinet, with integrated UPS

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SALES, SERVICE & SUPPORT

Baltic Scientific Instruments manufactures the analyzer platform. Commissioning and ongoing support are jointly provided by Fenno-Aurum Oy and Vocto Technologies Private Limited.

Function	Partner	Location	Contact
Manufacturer	Baltic Scientific Instruments	Riga, Latvia	www.bsi.lv
Authorised Distributor, Commissioning & Support	Fenno-Aurum Oy	Espoo, Finland	Esko Kantonen, CEO esko.kantonen@fennoaurum.com +358 40 356 4046
	Vocto Technologies Pvt Ltd	Chennai, India	Nithin Hari.T, Chief Revenue Officer Nithinhari@vocotechnologies.com +91 95006 54545
			Aswath.T, International Business Development Officer Aswath@voctotechnologies.com +91 78069 99078 Arul Jothi.R, Director-Technical Installation & Service +91 99941 78734 Sishaath R, USA Sales sishaath@voctotechnologies.com

REFERENCE INSTALLATIONS

On-line conveyor XRF systems installed worldwide, by country:

Country	Units	Country	Units
Austria	3	Kazakhstan	1
Belgium	5	Norway	1
Czech Republic	1	Russia	6
Finland	4	South Korea	2
France	2	Sweden	2
Germany	1	Zambia	4
Italy	1		



Fig. 7. Reference installations worldwide, with partner offices in Riga, Espoo and Chennai

PROVEN WORLDWIDE

33 Con-X systems installed across 13 countries

PARTNER WITH US

Contact Vocto Technologies (India) or Fenno-Aurum Oy (Finland) for a demonstration, site assessment, and calibration plan.

OUR GLOBAL PARTNERS

Design, manufacture & worldwide distribution

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Real-time, non-nuclear on-belt elemental grade control — engineered for the world's toughest plants.